

Kevin Xie

PHD CANDIDATE · PHYSICS · UNIVERSITY OF TORONTO

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PhD candidate in experimental quantum optics, operating at the intersection of quantum and classical theories. 8+ years of combined research experience in both academic and private sectors.

Education

University of Toronto

PHD PHYSICS

[Toronto, Canada](#)

Sep. 2020 - Present

University of Waterloo

BSC PHYSICS, CO-OP

[Waterloo, Canada](#)

Sep 2015 - Apr 2020

Honors & Awards

- 2025 **SGS Conference Travel Grant**, University of Toronto
- 2023 **Ontario Graduate Scholarship**, University of Toronto
- 2021 **Canada Graduate Research Scholarship – Master's**, NSERC
- 2020 **Dean's Honours List**, University of Waterloo
- 2017 **Undergraduate Student Research Award**, NSERC
- 2016 **President's Scholarship of Distinction**, University of Waterloo

[Toronto, Canada](#)

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Publications

* indicates shared first authorship.

PUBLISHED

- K. G. S. Xie***, C. J. Dale*, K. Pond Grehan, M. F. Wang, T. Enss, P. S. Julianne, Z. Yu, and J. H. Thywissen. Dimer-projection contact and the clock shift of a unitary Fermi gas. *Phys. Rev. Lett.* 136, 083402 (2026)
- C. J. Dale*, **K. G. S. Xie***, K. Pond Grehan, S. Zhang, J. Maki, and J. H. Thywissen. Emergent s-wave interactions in orbitally active quasi-two-dimensional Fermi gases. *Phys. Rev. A* 110, L051302 (2024)
- K. G. Jackson, C. J. Dale, J. Maki, **K. G. S. Xie**, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent s-wave interactions between identical fermions in quasi-one-dimensional geometries. *Phys. Rev. X* 13, 021013 (2023)
- C. R. C. Buhariwalla, Q. Ma, L. DeBeer-Schmitt, **K. G. S. Xie**, D. Pomaranski, J. Gaudet, T. J. Munsie, H. A. Dabkowska, J. B. Kycia, and B. D. Gaulin. Long-wavelength correlations in ferromagnetic titanate pyrochlores as revealed by small-angle neutron scattering. *Phys. Rev. B* 97, 224401 (2018)

Presentations

CONTRIBUTED TALKS AND POSTERS

- K. G. S. Xie**, C. J. Dale, K. Pond Grehan, M. F. Wang, Z. Yu, P. S. Julianne, T. Enss, and J. H. Thywissen. Clock shift and contact of a unitary Fermi gas inferred by dimer association. Oral presentation: DAMOP 2025, Portland, Oregon, U.S.A.
- K. G. S. Xie**, C. J. Dale, K. Pond Grehan, M. F. Wang, Z. Yu, P. S. Julianne, T. Enss, and J. H. Thywissen. In search of C: the need for speed. Oral presentation: TUCAN 2025, Waterloo, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Emergent s-wave dimers near a p -wave Feshbach resonance in a strongly confined Fermi gas. Oral presentation and poster: ICPEAC 2023, Ottawa, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, K. Pond Grehan, J. Maki, S. Zhang, and J. H. Thywissen. Associating emergent s-wave dimers along strongly confined directions of a spin-polarized Fermi gas. Oral presentation: DAMOP 2023, Spokane, Washington, U.S.A.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Zhang, and J. H. Thywissen. Feshbach-enhanced odd- and even-wave interactions between identical fermions in quasi-one-dimensional confinement. Poster: ICAP 2022, Toronto, Ontario, Canada.
- K. G. S. Xie**, K. G. Jackson, C. J. Dale, J. Maki, B. A. Olsen, D. J. M. Ahmed-Braun, S. Kokkelmans, S. Smale, P. S. Julianne, and J. H. Thywissen. P -wave interactions in a spin-polarized Fermi gas with and without confinement. Poster: DAMOP 2022, Orlando, Florida, U.S.A.

Teaching

2026	PHY385 , Introductory Optics, Teaching Assistant	Toronto, Canada
2025	PHY324 , Practical Physics II, Teaching Assistant	Toronto, Canada
2025	PHY293 , Waves and Modern Physics, Teaching Assistant	Toronto, Canada
2024	PHY294 , Quantum and Thermal Physics, Teaching Assistant	Toronto, Canada
2024	PHY405 , Electronics, Teaching Assistant	Toronto, Canada
2023	PHY424 , Advanced Physics Laboratory, Teaching Assistant	Toronto, Canada
2021, 2022	PHY224 , Practical Physics I, Teaching Assistant	Toronto, Canada
2020, 2021	PHY131 , Introduction to Physics I, Teaching Assistant	Toronto, Canada

Academic Services

The 9th Annual Meeting of the Toronto Ultracold Atoms Network (TUCAN 2026)

[Toronto, Canada](#)

ORGANIZER

May 2026

- Primary organizer of provincial meeting for ultracold atom research with 75+ attendees. Coordinated programming, logistics, and communications.

Reviewer for Phys. Rev. X

REFEREE

2024

University of Toronto Graduate Students' Union

[Toronto, Canada](#)

MEMBER OF BOARD OF DIRECTORS AND ELECTIONS & REFERENDA COMMITTEE

Oct. 2023 – Apr. 2024

- Organized and supported policy development and governance initiatives affecting 20,000+ graduate students, contributing to improvements in academic, financial, and community resources.

The 27th International Conference on Atomic Physics (ICAP 2022)

[Toronto, Canada](#)

VOLUNTEER

July 2022

- Assisted in organizing and running an international conference with 500+ participants, providing on-site logistics, A/V support, and management of virtual conference components.

Experiences

University of Toronto (Prof. Joseph Thywissen)

[Toronto, Canada](#)

PHD RESEARCHER, EXPERIMENTAL QUANTUM OPTICS

May 2020 – Present

- **Research focus:** precision measurements, high-dimensional data analysis, control of complex physical systems.
- Conducted experiments on ultracold quantum gases, including spectroscopy of strongly interacting fermions to study many-body phenomena.
- Applied model fitting, parameter estimation, and uncertainty analysis to extract physical parameters from noisy data.
- Deployed and maintained custom control, real-time monitoring, and automated data-acquisition systems (Python, C++).
- Led a research team of four, directing experimental strategy, coordinating measurement protocols, and managing data-analysis pipelines.
- Regularly communicated results to international collaborators; presented findings at conferences and peer-reviewed journals.

University of Waterloo (Prof. Jan Kycia)

[Waterloo, Canada](#)

UNDERGRADUATE RESEARCHER

Jan. – Sep. 2017, Sep. 2019 – Apr. 2020

- Engineered and analyzed thermal properties of components for a hybrid Helium-3 dilution refrigerator aimed at millikelvin-scale spin-qubit characterization.

Lumentum

[Ottawa, Canada](#)

OPTICAL ALGORITHMS DEVELOPER

Sep. – Dec. 2018, May – Aug. 2019

- Built a predictive failure-detection neural network model (>96% accuracy) to optimize calibration of LCoS components in optical networking devices, aimed at improving throughput and automation.
- Standardized test-station infrastructure across multiple facilities, eliminating bottlenecks and increasing throughput by up to 10%.
- Conducted comprehensive performance studies on next-generation bandwidth-enhancement algorithms for optical module R&D.
- Replaced outdated internal component data retrieval tool with modern Python-based solution adopted company-wide.

TRIUMF (T2K Canada)

[Vancouver, Canada](#)

UNDERGRADUATE RESEARCHER

Sep. 2017 - Apr. 2018

- Detected and corrected periodic temperature fluctuations that introduced systematic error in photomultiplier-tube characterization. Validated and reproduced by international collaborators (KEK).

Skills

Programming Languages	Python (NumPy, SciPy, Pandas), MATLAB, Mathematica, C++, Linux, Git, LaTeX
	English (native), Cantonese (beginner/intermediate)